

CAPO: Compute-Aware Automated Protein-Model Optimization from Raw Data to Fine-Tuned Models

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CAPO turns raw biological assay data into fine-tuned protein models through auditable, leakage-aware, budget-aware decisions.

What is CAPO?

An **agentic system** that takes raw experimental protein data all the way to fine-tuned, evaluated models, with the full path from reads to results kept auditable.

CAPO makes costly choices **explicit**: data filtering, leakage control, model and hardware selection, training strategy and the run's **actual cost**, each logged and auditable.

Motivation

Protein-model adaptation is a **manual, error-prone bottleneck**: data, evaluation and compute must line up and one wrong call can silently waste GPU hours or invalidate results.

- **Data**: inputs, labels and metadata can misalign or drift.
- **Evaluation**: leakage and imbalance inflate scores.
- **Compute**: invalid runs burn GPU hours before failing.

Existing agents plan, write and run code but skip these checks. **CAPO** adds the missing **run-level gate: biological validity, schema integrity and compute feasibility**. It also **matches the model to the task**, choosing from 30+ candidates.

Contributions

- **End-to-end** adaptation in one auditable workflow.
- **Leakage-aware** dataset preparation & evaluation splits.
- **Compute-gated**: data, code, hardware and cost checked *before* GPU training.

Next steps

Extending **CAPO** to three open-access protein tasks: variant-effect prediction, peptide-MHC binding and protein-ligand affinity, with ablations measuring each component's impact on performance and cost.

How CAPO works

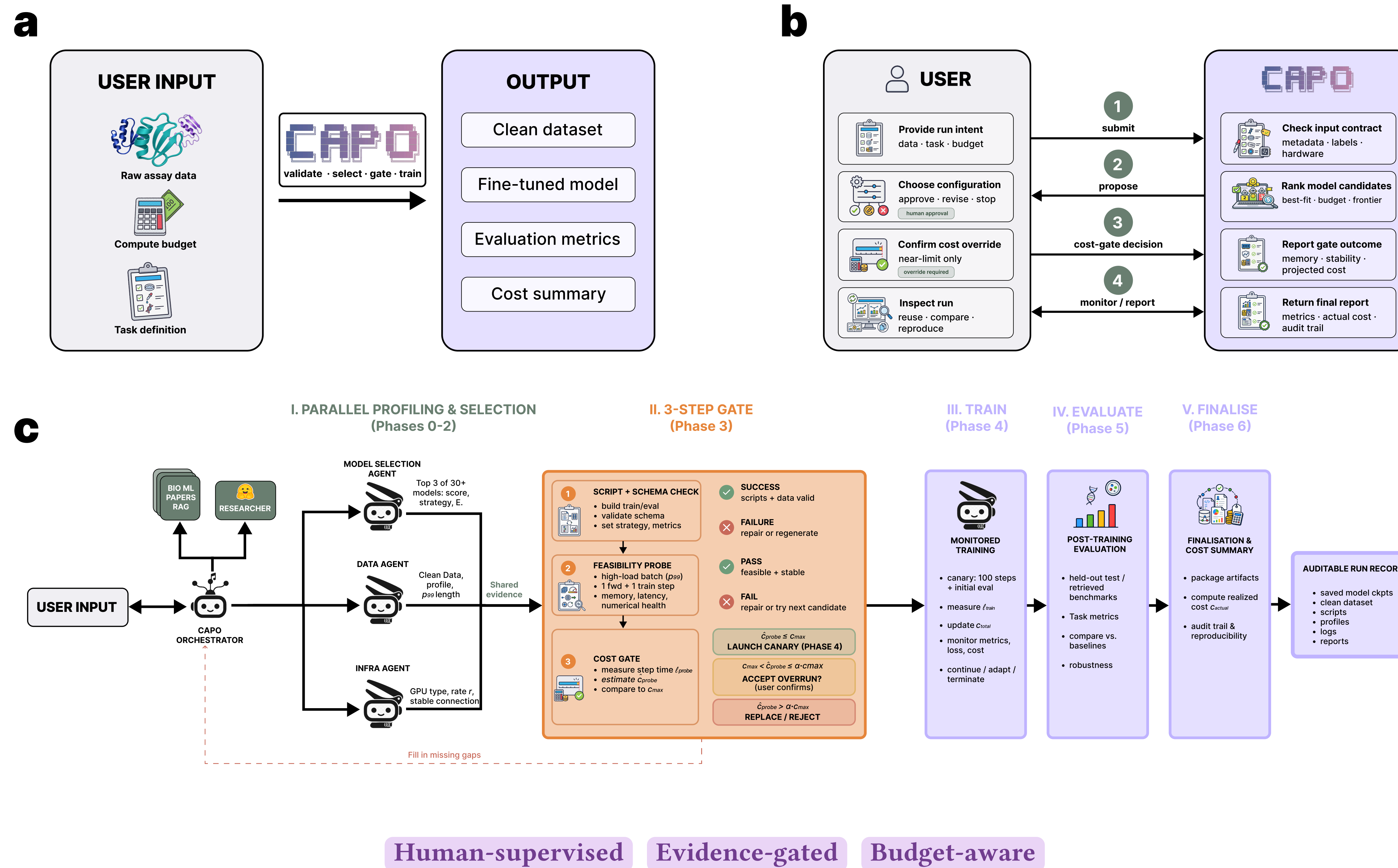
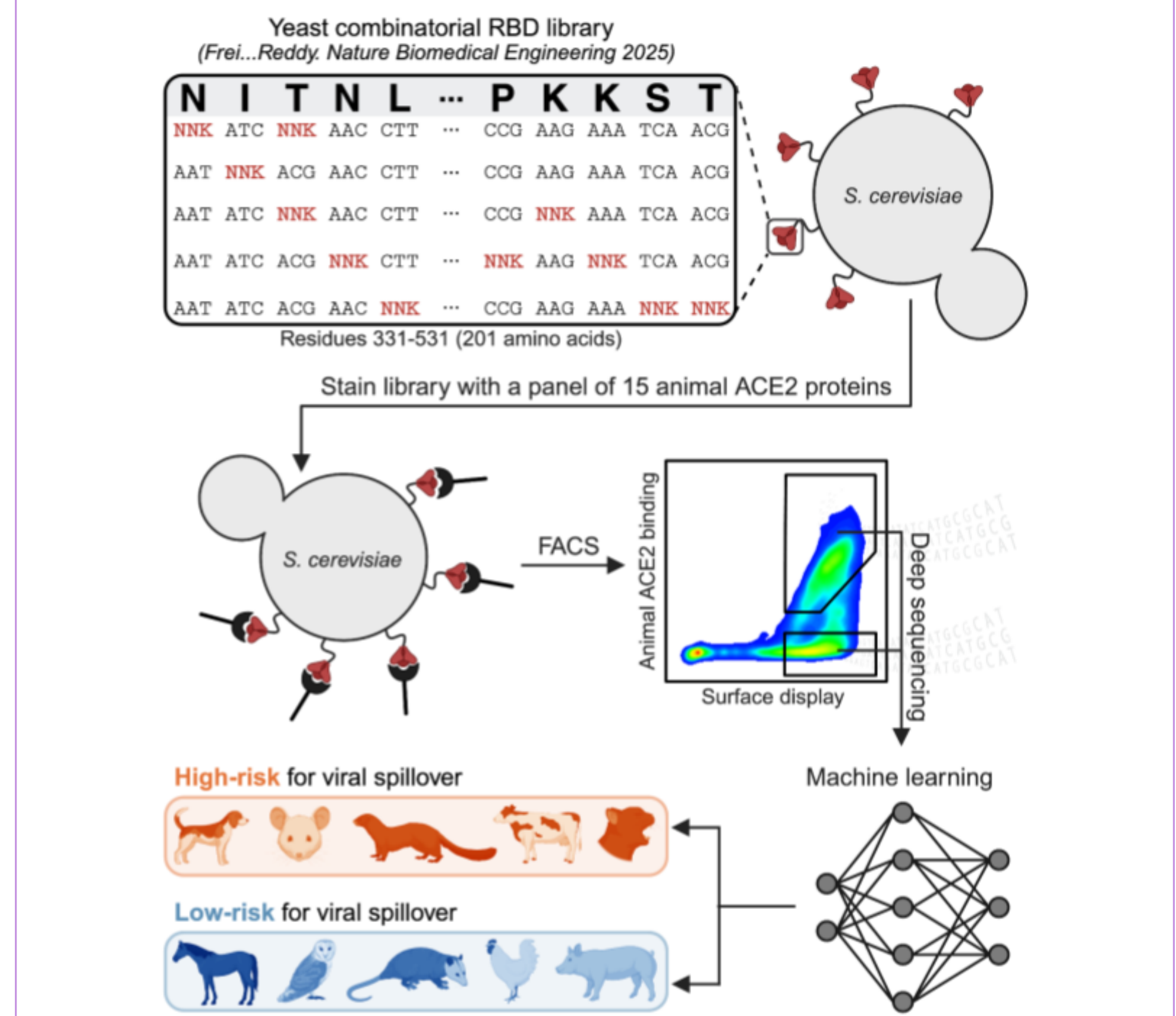


Figure 1: (a) Inputs (intent, data, budget) → outputs (clean dataset, fine-tuned model, metrics, cost). (b) Human/agent loop. (c) Agents profile data, rank 3 of 30+ models and resolve infrastructure. A **three-step gate** (script+schema → feasibility → cost) admits a run before monitored training, evaluation and finalisation.

Dataset & evaluation



RBD-ACE2 binding. A yeast-displayed SARS-CoV-2 Omicron BA.1 RBD mutagenesis library (residues 331–531) is stained against **10 animal ACE2 orthologs**, sorted by FACS and deep-sequenced. Task: predict binding per (RBD, ortholog) pair. Metric: **MCC**.

Dataset available from the authors upon request.

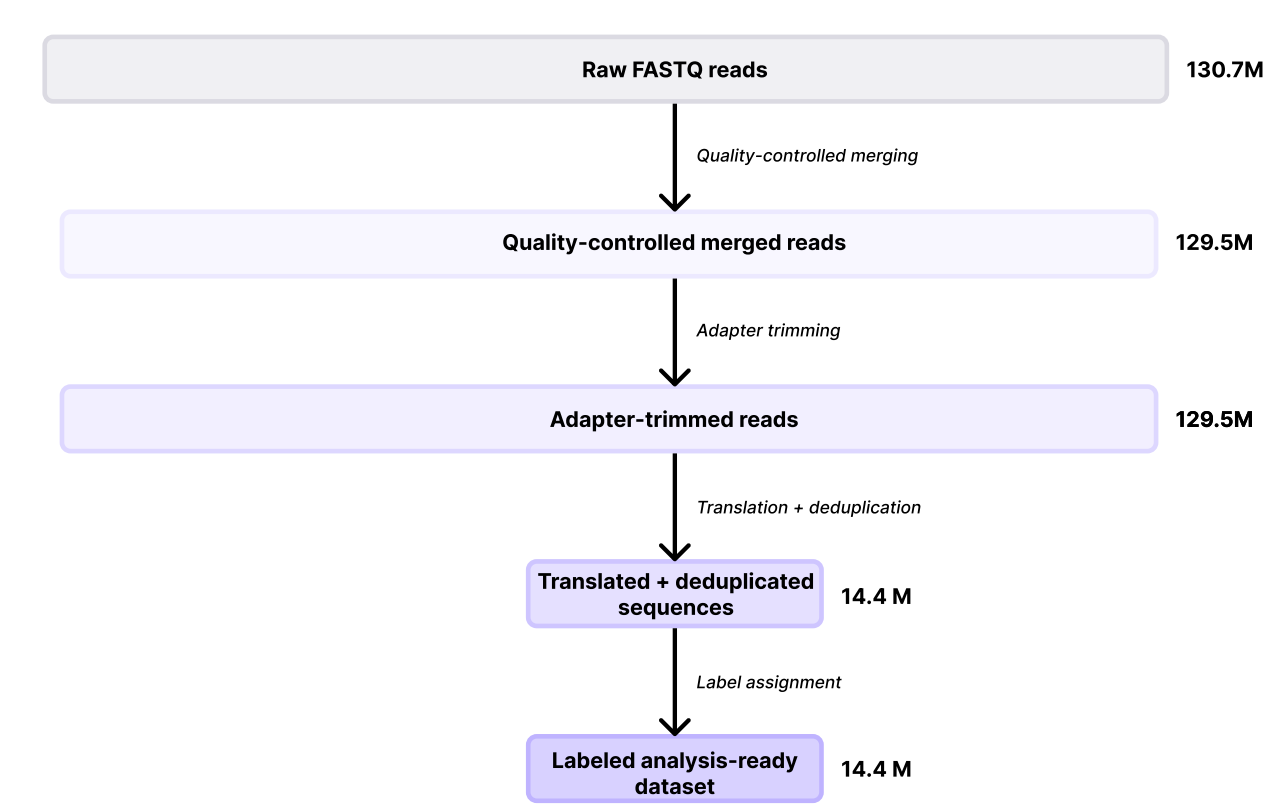
Code & models



References

[1] Schmirler et al. Fine-tuning PLMs boosts predictions. *Nat. Commun.* 2024. [2] Bikias et al. PLMFit. *Brief. Bioinform.* 2025. [3] Liu et al. AutoProteinEngine. *COLING* 2025. [4] Bran et al. ChemCrow. *Nat. Mach. Intell.* 2024. [5] Lin et al. VarPred: missense pathogenicity via PLMs. *Sci. Rep.* 2024. [6] Wohlwend et al. MUNIS-HLA-1 CD8* T-cell epitopes. *Nat. Mach. Intell.* 2025. [7] Affinity fine-tuning of Boltz-2 for protein-ligand binding. *bioRxiv* 2026.

From raw reads to clean data



CAPO turns raw paired-end sequencing libraries into **ML-ready** protein sequences with **expert-level** annotation, applying quality control, adapter trimming, filtering and deduplication to remove technical noise while preserving biological signal.

A combinatorial SARS-CoV-2 receptor binding domain (RBD) library ($\sim 2 \times 10^8$ variants) is FACS-sorted for ACE2 binding across **10 animal species**, then sequenced on a **next-generation sequencing** (NGS) platform. From the raw NGS data, **CAPO** discards low-quality reads and rare variants (<2 counts) and translates to amino acids, yielding clean, **multilabel** training sets that annotate each variant across all 10 species, enabling prediction of cross-species binding and **reverse-zoonotic** spillover risk.

System	Output	Match	Cost
<i>Pangolin host</i>			
CAPO	full-length canonical RBDs (14.4M)	97.6%	\$9.91
gen. agent read-window fragments		0.0%	\$25.96
<i>Possum host</i>			
CAPO	canonical 201-aa RBDs (2.61M)	99.83%	\$6.88
gen. agent short fragments (79–167 aa)		0.00%	\$7.09

"Match" = agreement with an independently curated gold set of full-length RBDs. The general agent stops at read merging, so its fragments cannot match the gold sequences.

Results

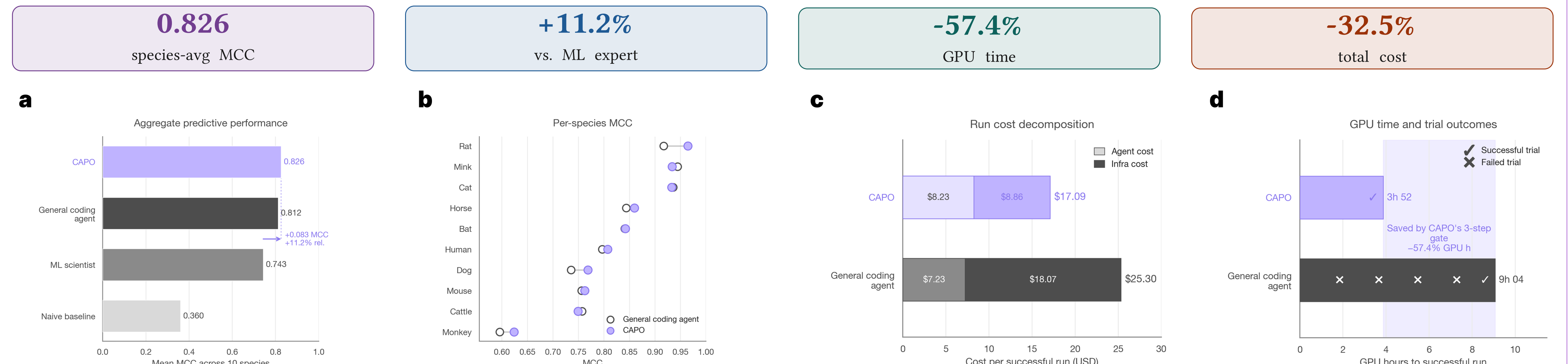


Figure 2: CAPO beats a naive baseline, an ML-expert pipeline and a general coding agent on (a) aggregate and (b) per-species MCC. Its pre-launch gate eliminates failed GPU trials, (c) cutting cost per successful run from \$25.30 to \$17.09 and (d) GPU time by 57.4%.